

Analysis of the Complement System in the Clinical Immunology Laboratory



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KEYWORDS

- Complement • CH50 • C3 • C4 • Classical pathway • Lectin pathway
- Alternative pathway • Membrane attack complex

KEY POINTS

- The initial steps in the complement cascade are triggered by (1) antigen-antibody complexes (classical pathway), (2) lectin binding (lectin pathway), or (3) accelerated C3 tick-over (alternative pathway).
- Formation of the C3 and C5 convertases results in production of anaphylatoxins (C3a and C5a) and the downstream formation of the membrane attack complex.
- Complement receptors are G-protein-coupled receptors expressed on various immune cells and rarely on other cells. These receptors are activated by complement protein fragments that are generated in the plasma during complement activation.
- Fluid phase and membrane-bound regulators promote effective host defense against infection and immune homeostasis and prevent inappropriate complement activation.
- Functional complement activity and quantitation of individual complement components facilitates the diagnosis of diseases of complement deficiency, exuberant complement activation, and complement dysregulation.

HISTORY AND INTRODUCTION

The discovery of complement dates to the late 1800s, when Jules Bordet identified an unknown heat-labile lytic substance in blood that possessed activity against various forms of bacteria. This heat-labile mediator appeared to promote nonspecific antimicrobial activity, complementing the function of heat-stable serum proteins

Disclosure Statement: The authors have nothing to disclose.

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Clin Lab Med 39 (2019) 579–590

<https://doi.org/10.1016/j.cl.2019.07.006>

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(ie, antibodies). This component was thus termed "complement" by Paul Ehrlich in 1899.¹ The complement system is now known to be a key component of the innate immune system. Activation of complement involves a complex cascade of proteases, providing a first line of defense against pathogens as well as playing a role in eliminating cellular debris. Its function complements that of antibodies and phagocytes by (1) eliminating microbes (through cytolytic membrane attack by disrupting bacterial cell walls), (2) removing apoptotic cells and debris (through opsonization of antigens to promote phagocytosis), (3) augmenting inflammation (through chemoattraction and activation of leukocytes, particularly macrophages and neutrophils), and (4) acting as costimulators of B-cell activation and antibody production and thereby augmenting innate and adaptive immune responses (eg, via C3d).² Normal host (self) cells express regulatory proteins that protect against complement-mediated attack, whereas foreign pathogens (non-self) and altered host cells are susceptible to such attack, leading to opsonization of pathogens and autoreactive cells.

Soluble proteins of the complement system are synthesized by the liver and circulate initially as inactive pro-enzymes that become functionally active on cleavage by a protease. In the setting of infection or inflammation, the production of certain complement proteins and other acute phase reactants is augmented by inflammatory cytokines, such as tumor necrosis factor α , interleukin (IL)-1, and IL-6, which are produced by leukocytes (eg, macrophages and neutrophils).² The initial steps in the complement cascade are triggered by (1) antigen-antibody complexes (*classical pathway*), (2) lectin binding (*lectin pathway*), or (3) accelerated C3 tick-over (*alternative pathway*) (Fig. 1). After this initial activation, protease-mediated cleavage of complement proteins leads to activation of the aforementioned complement-mediated functions and downstream activation of adaptive immune responses. The other role for complement is immune homeostasis as clearance of apoptotic cells and debris removes self-antigens from the circulation, which reduces the risk of developing autoimmunity.

CLASSICAL PATHWAY

The classical pathway is initiated by C1q. When C1q recognizes certain target molecules (eg, immune complexes or surface-bound pentraxins), a conformational change occurs, allowing C1q to interact with C1r and C1s and form activated C1qr₂s₂.³ The active enzymatic site is on C1s. Subsequently, activated C1qr₂s₂ cleaves C4, then C2, to generate the classical pathway C3 convertase, C4b2a. C4b2a is then able to induce proteolytic activity and activate the common terminal pathway.⁴

LECTIN PATHWAY

The lectin pathway can be triggered by circulating pattern recognition receptors (PRRs) that recognize carbohydrates on microbial surfaces. These include (1) mannose binding lectin (MBL), a "collectin"; and (2) ficolins. Ficolins are PRRs with a fibrinogen-like domain, for example, M-ficolin (ficolin-1, secreted by lung and blood cells), L-ficolin (ficolin-2), and H-ficolin (ficolin-3); the latter two are synthesized in the liver and circulate in blood. MBL-associated serine proteases (MASP-1 and MASP-2) are evolutionarily related to C1r and C1s, and function similarly.⁵ MBL complexes with MASPs, which then cleave C4 and C2 to form the lectin pathway C3 convertase, C4b2a.

ALTERNATIVE PATHWAY

The alternative pathway is activated by augmentation of C3 tick-over, which is the spontaneous hydrolysis of the thioester bond of C3 that occurs at a low level

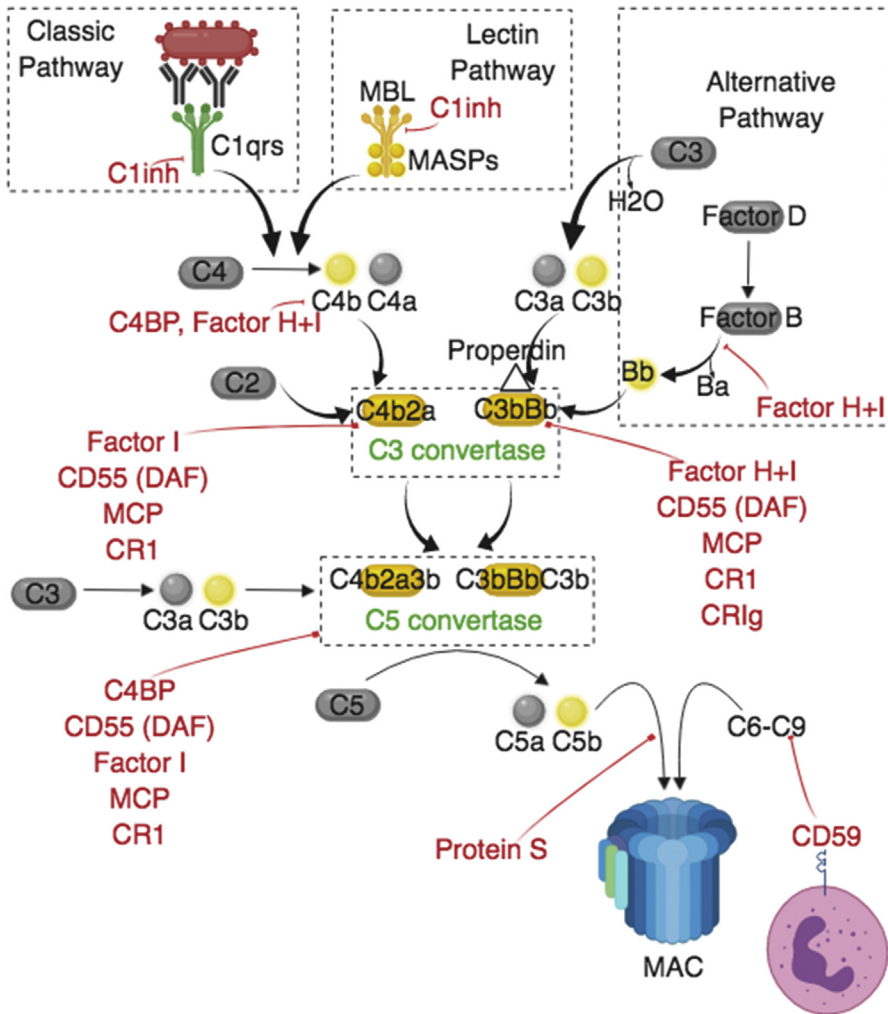


Fig. 1. Overview of the complement system and regulatory proteins. Activation occurs through 3 different pathways: the classical, lectin, and alternative pathways. The classical pathway is initiated through the activated C1 complex (C1qrs), and the lectin pathway is triggered by a number of PRRs, including MBL and MASPs 1 and 2. The alternative pathway can be activated by spontaneous hydrolysis of C3 (tick-over) or by C3 formed via the classical and lectin pathways. Activation of complement results in the production of the classical and lectin pathway C3 convertase (C4b2a) and the alternative pathway C3 convertase (C3bBb), which have proteolytic activity for C3 and form their respective C5 convertases by generation of more C3b (C4b2a3b and C3bBbC3b). Cleavage of C5 leads onward to the common terminal pathway, which produces the cytolytic MAC. Regulatory proteins are present both in the fluid phase or bound to the cell surface. The major fluid phase regulators include the serum C1INH, C4BP, protein S, factor H, factor I, and an AI, whereas the major cell-bound regulators include CD55 (DAF), CD59 (MAC-IP, or protectin), MCP (or CD46), CR1 (or CD35), and CRIg. (Figure created with BioRender.)

constitutively, but does not proceed further if C3b does not encounter its stabilizing counterpart, Bb.^{6,7} Bb is a proteolytic fragment formed as a result of factor D-dependent cleavage of factor B. The alternative pathway C3 convertase, C3bBb, is formed when C3b binds to Bb. The activity of C3bBb is stabilized by properdin, which is found on activated surfaces with decreased sialic acid content (eg, foreign cell membranes), but not on host cell membranes rich in sialic acid. The labile thioester group on C3b is also able to covalently and stably bind exposed amino or hydroxyl groups on the neighboring cell surfaces of pathogens or altered host cell membranes. This process augments C3 tick-over while also promoting subsequent complement activation on the surfaces of targeted pathogens or altered self-structures.⁸ The alternative pathway can also be activated or augmented by C3b generated by the classical or lectin pathways.

COMMON TERMINAL PATHWAY

After formation of the C3 convertase, C4b2a via the classical and/or lectin pathways, or C3bBb via the alternative pathway, the complement cascade proceeds due to the proteolytic activity of the C3 convertase on C3, which produces an inflammatory mediator or “anaphylatoxin” (C3a) and an enzymatic cleavage component (C3b). This leads to the formation of their respective C5 convertases (C4b2a3b and C3bBbC3b).³ Subsequent cleavage of C5 by the C5 convertase also produces an inflammatory mediator or “anaphylatoxin” (C5a) and an enzymatic cleavage component (C5b). The anaphylatoxins C3a, C4a, and C5a activate mast cells and the resulting mediators contribute to the vascular phase of inflammation. C5a is also a chemoattractant that recruits neutrophils and monocytes, contributing to the cellular phase of inflammation. C5b initiates the common terminal pathway and, after complexing with C6, C7, C8, and C9, forms the cytolytic membrane attack complex (MAC).

Complement receptors are G-protein-coupled receptors expressed on various immune cells and rarely on other cells (Table 1). These receptors are activated by complement protein fragments, principally those derived from C3, which are generated in the plasma during complement activation.⁹ C3a is recognized by C3aR, which is expressed on mast cells and basophils and functions as a potent anaphylatoxin. The ligand for complement receptor 1 (CR1) is C3b, which is expressed on red blood

Table 1 Complement receptors			
Ligand	Receptor	Cells of Distribution	Function
C3a ^a	C3aR	Mast cells, basophils	Anaphylatoxin
C3b ^a	CR1 (CD35)	RBCs, WBCs, monocytes, DCs, podocytes	Immune complex clearance, phagocytosis
C3d ^a	CR2 (CD21)	B cells	Costimulation
iC3b ^a	CR3 (CD11b/CD18)	Phagocytic cells	Phagocytosis
C3dg ^a	CR4 (CD11c/CD18)	?NK cells, monocytes, DCs	?Phagocytosis
C5a	C5aR (CD88)	WBCs, monocytes, DCs, certain tissues	Anaphylatoxin
C5a	C5L2 (intracellular, not functionally a GPCR)	WBCs, monocytes, DCs, certain tissues	?Decoy receptor

Abbreviations: DC, dendritic cell; GPCR, G protein-coupled receptor; NK, natural killer; RBC, red blood cell; WBC, white blood cell.
^a Fragments of C3.

cells (RBCs), white blood cells (WBCs), monocytes, dendritic cells (DCs), and podocytes. Its functions include immune complex clearance and phagocytosis. The ligand for CR2 is C3d, which is expressed on B cells and functions in costimulation. The ligand for CR3 is iC3b, which is expressed on phagocytic cells and is important in phagocytosis. Finally, CR4 recognizes C3dg, which is expressed on natural killer cells, monocytes, and DCs and may play a role on phagocytosis as well. **Table 1** also summarizes the role of C5a receptors.¹⁰

FLUID PHASE REGULATORS

To promote effective host defense against infection and immune homeostasis and to prevent inappropriate complement activation, regulatory proteins play an important role, whether present in the fluid phase or bound to the cell surface. The major fluid phase regulators include the serum C1 inhibitor (C1INH), C4 binding protein (C4BP), protein S, factor H, factor I, and an anaphylatoxin inhibitor (AI).¹¹

C1 INH is a member of the serpin family of serine protease inhibitors and is an important fluid phase regulator as it inhibits several circulating complement proteases (C1r, C1s, and the MASPs), contact system proteases (HMW kininogen, kallikrein, factor XIa, and factor XIIa), and fibrinolytics (plasmin).¹² When these proteolytic cascades are activated by C1q, bradykinin (derived from contact activation of factor XII) and several other vasoactive substances are generated, resulting in increased vascular permeability.

C4BP binds and degrades C4b to form C4c and C4d.¹³ C4d does not have protease activity; rather, its production limits amplification of complement activation via the classical and lectin pathways.

Protein S, a cofactor for activated protein C (APC) in the coagulation cascade, binds to the C5bC6C7 complex to prevent insertion into the cell membrane and subsequent formation of a cytolytic complex with C8 and C9.¹⁴ Protein S can also bind C4BP to influence the coagulation cascade, which highlights the close association between the coagulation and complement systems.¹⁵

Serum regulatory protein factor H is able to bind C3b and interfere with the ability of C3b to cleave C5 and factor B on cell surfaces, which would in turn inhibit the formation of C3 and C5 convertases. Factor H also acts as an essential cofactor for factor I, another fluid phase regulator that mediates proteolytic degradation of plasma C3b. The importance of factor H is highlighted by disorders that arise from certain mutations or single nucleotide polymorphisms in the gene, including age-related macular degeneration (AMD) and atypical hemolytic uremic syndrome (aHUS).¹⁶

MEMBRANE-BOUND REGULATORS

The major cell-bound regulators include CD55 (delay accelerating factor, or DAF), CD59 (MAC-inhibitory protein, or MAC-IP, or protectin), membrane cofactor protein (MCP, or CD46), CR1 (or CD35), and CR1g (complement receptor of the immunoglobulin superfamily).¹¹

CD55 (or DAF) is a membrane-bound glycoprotein that promotes the decay dissociation of C4b2a and thus interrupts formation of the classical and lectin pathway C3 convertases. By limiting C3b formation, CD55 also inhibits signal amplification by C3b via the alternative pathway.¹⁷

CD59 is a membrane-bound glycoprotein that can hinder the polymerization of C9 and thereby inhibit the formation of the MAC and cytotoxicity.¹⁸

Similar to factor H, MCP (CD46) can serve as a cofactor for factor I-mediated proteolysis of plasma C3b.¹⁹

Membrane-bound CR1 is a complement receptor that also binds to C3b in immune complexes and serves as a cofactor for factor I-mediated degradation of C3b.

CR1g is a complement receptor on tissue-resident macrophages that recognizes C3b-opsonized surfaces and plays a role in immune clearance. However, the extracellular domain of CR1g binds to C3b in such a way that it interferes with substrate binding, limiting C3b-mediated activation of C3 and C5, thus inhibiting the alternative pathway.²⁰

LABORATORY TESTING

Complement assays are performed on serum samples received in gold or red top tubes transported on ice to limit ex vivo tick-over. Improper handling and/or prolonged transport time at room temperature may affect the interpretation of antigenic and functional assays. On receipt, the sample is centrifuged and the serum is aliquoted and stored at -70°C .

Methods of Complement Testing

Functional complement activity is traditionally assessed using a hemolytic assay. To evaluate classical pathway activation, serial dilutions of the sample are made and incubated with antibody-sensitized sheep erythrocytes. The CH50 is the reciprocal of the dilution at which 50% hemolysis is seen. Similarly, alternative pathway activation can be assessed using rabbit (or guinea pig) erythrocytes, which specifically activate the alternative pathway due to decreased sialic acid content. The AH50 is the reciprocal dilution at which 50% hemolysis is seen. Alternatively, quantitation of classical pathway activation can be performed using nephelometric, turbidimetric, or enzyme-linked immunosorbent assay–based assays of formation of the neoantigen MAC complex, which is a less time-consuming assay that can be automated, and has been demonstrated to have excellent concordance with the hemolytic assay.

Individual complement components, quantitated in the past by immunodiffusion, is now measured by nephelometry or turbidometry,²¹ with the caveat that the selection of antibodies may affect the interpretation of the assay depending on what epitopes are recognized by the antibodies (ie, intact protein only, soluble fragments after proteolysis only, or both). Most assays for C3 and C4 quantitate intact proteins as well as the major soluble fragments formed during activation. However, interpretation may be complicated by limited detection of certain proteolytic products. Membrane-bound regulatory proteins such as CD55, CD59, and CD35 (CR1) can be assessed using flow cytometry.²²

Diagnostic Considerations and Laboratory Testing for Complement Deficiency

Fig. 2 highlights a suggested algorithm for determining the possible cause of complement deficiency. If CH50 is low and AH50 is normal, this finding points to a likely deficiency in a complement protein or regulatory protein involving the classical pathway. Factors that should be considered for quantitative and functional testing include C1, C2, and C4. C1 inhibitor deficiency evaluation should be considered in patients with nonpruritic angioedema without urticaria. If a single component is abnormal, the defect may either be hereditary or functional; genetic testing may be helpful to sort this out. However, if there are abnormalities in multiple components of the classical pathway, the problem is likely one of complement consumption as can be seen in immune complex diseases such as systemic lupus erythematosus (SLE), Sjögren's syndrome, rheumatoid arthritis, and cryoglobulinemia.

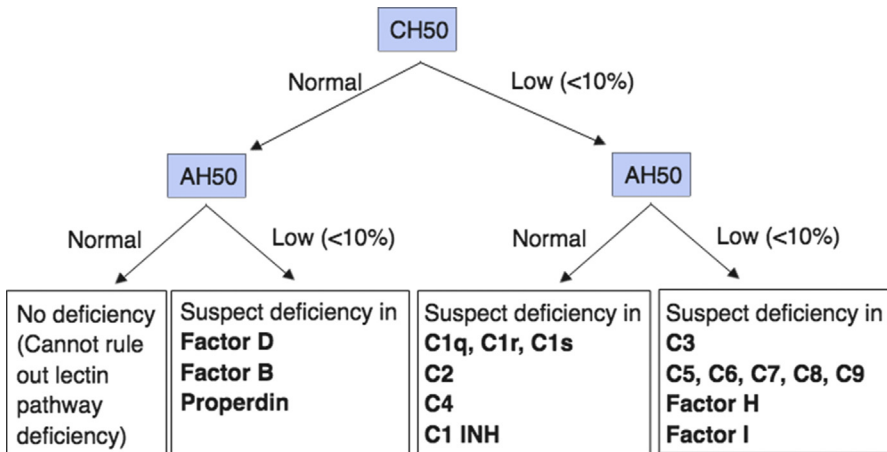


Fig. 2. Algorithm for laboratory testing of complement deficiency. When a disorder of complement is suspected, measurement of CH50 and AH50 is a recommended starting point for laboratory testing of complement deficiency and dysregulation. Antigenic or functional deficiency can be confirmed through the addition of individual purified components with correction of CH50 or AH50. When a particular factor is implicated, it may be appropriate to assess antigen levels of the factor as well as associated regulatory proteins to differentiate antigenic deficiency from functional deficiency.

If CH50 is normal and AH50 is low, this is consistent with a likely deficiency in a complement protein or regulatory protein involving the alternative pathway. Factors that should be considered for quantitative and functional testing include properdin, and antigen levels should be considered for factor B, factor D, factor H, and factor I, and MCP. Deficiencies of properdin and factor B/D are associated with Neisserial and pneumococcal infections. Factor H/I and MCP haploinsufficiency are seen in aHUS. Factor H abnormalities have also been associated with membranoproliferative glomerulonephritis and AMD.^{23,24} Abnormalities may indicate a hereditary deficiency of one or more of these regulatory factors. Rarely, gram-negative sepsis, Henoch-Schönlein purpura, IgA nephropathy, and complement factor-mediated glomerulonephritis (C3 nephritic factor) may be associated with a decrease in alternative pathway activation (low AH50) with preservation of normal classical pathway function (normal CH50). C3 nephritic factor is an immunoglobulin (Ig)G antibody that stabilizes C3bBb, promoting C5 activation and MAC-mediated glomerulonephritis (aka type II membranoproliferative glomerulonephritis).

If CH50 and AH50 are both low, this is consistent with a likely deficiency in a complement protein or regulatory protein involving the terminal pathway. Factors that should be considered for quantitative and functional testing include C3, C5, C6, C7, C8, and C9, and antigen levels should be considered for factor H and factor I. If a single component is abnormal, the deficiency may be hereditary, and genetic testing may be helpful. However, if there are abnormalities in multiple components of the terminal pathway, the problem is likely one of complement consumption (eg, SLE, other immune complex disease, or shock).

If CH50 and AH50 are both normal but there is high clinical suspicion for complement deficiency, the defect may be in the lectin pathway. In this case, measurement of MBL may be helpful. However, it is unlikely that a systemic disease causing complement consumption would specifically target the lectin pathway while leaving the classical and alternative pathways intact.

Diagnostic Considerations and Laboratory Testing for Complement Dysregulation

Table 2 highlights laboratory test patterns of complement activation in systemic disease. Activation of the classical pathway would typically manifest as low CH50, low C3, low C4, and normal factor B. Relevant conditions in which such a pattern could be seen include warm autoimmune hemolytic anemia, SLE, Sjögren syndrome, rheumatoid arthritis, and cryoglobulinemia.

With regard to the alternative pathway, activation would typically manifest as low CH50, low C3, normal C4, and low factor B. Relevant conditions in which such a pattern could be seen include aHUS and C3 nephritic factor glomerulonephritis (C3 NF GN). Typical hemolytic uremic syndrome (HUS) presents as a clinical triad of nonimmune hemolytic anemia, thrombocytopenia, and renal impairment, but in the absence of a known bacterial trigger, this syndrome is termed aHUS, which comprises approximately 10% of total cases and has a poor prognosis.^{25–27} Approximately 50% to 60% of familial and sporadic aHUS cases are linked to various mutations of the complement pathway.^{28,29} The etiology of aHUS is quite heterogeneous, as complete or partial factor H deficiency (the most common),^{30–32} deficiency in MCP (CD46) (second most common, seen in 10%–15% of patients with aHUS),^{33–35} and many other mutations have been described (for example, gain of function mutations of factor B and C3, and loss of function mutations of factor I, MCP, thrombomodulin, and CFHR1/3 [complement factor H-related pseudogenes]).^{33,36–38} Patients with aHUS exhibit low C3, high levels of activated complement components (C3b, C3c, C3d), and normal C4, which suggests selective activation of the alternative pathway.³⁰ The C3 convertases of the classical and lectin pathways are formed by C2 and C4 proteolysis, whereas the alternative pathway convertase cleaves C3 and bypasses C4.³⁹

Activation of the terminal pathway would typically manifest as low CH50, low C3, normal C4, and normal factor B. This can be seen in conditions such as paroxysmal nocturnal hemoglobinuria (PNH). Patients with PNH present with the clinical triad of hemolytic anemia, bone marrow failure, and thrombosis.⁴⁰ On a baseline of chronic low-level hemolysis, patients often experience episodic severe hemolysis triggered by perturbations, such as surgery, infection, or inflammation. PNH arises from an expanded clonal proliferation of hematopoietic cells with somatic mutations in phosphatidylinositol glycan class A (PIGA), a gene involved in glycosylphosphatidylinositol (GPI) anchor biosynthesis.^{41–44} Both CD55 (DAF) and CD59 (MAC-IP) are GPI-anchored proteins with a crucial role in complement regulation. Global deficiency of CD55 and CD59 results in dysregulated complement activation, leading to complement-mediated intravascular hemolysis of erythrocytes as well as activation of other hematopoietic cells. PNH is a clinical diagnosis (one may observe intravascular hemolysis, thrombosis, bleeding diathesis, and/or recurrent infections) that can be confirmed by findings of low CH50 and low AH50, flow cytometry

Table 2				
Laboratory test patterns of complement activation				
Profiles of Activation	CH50	C3	C4	Factor B
Classical	↓	↓	↓	Normal
Alternative	↓	↓	Normal	↓
Terminal	↓	↓	Normal	Normal
Fluid phase (classical)	↓	Normal	↓	Normal
Acute phase	↑	↑	↑	↑

↑ (or ↓) indicate increase (or decrease) of the indicated complement component or factor.

demonstrating CD55 and CD59 deficiency, and genetic testing (if needed). Although the Ham test (which involves assessing RBC fragility in a mild acid) was once used for this condition, it is now considered obsolete due to low sensitivity and specificity. Eculizumab is a promising therapeutic for the treatment of both aHUS and PNH; it specifically binds to the terminal complement component C5, inhibiting its cleavage by the C5 convertase.⁴⁵ Due to the potent effect of eculizumab in blocking MAC formation, meningococcal vaccination is imperative to reduce the risk of invasive meningococcal disease, though even this measure does not completely mitigate the risk.⁴⁶

Activation of the fluid phase components of the classical pathway would typically manifest as low CH50, normal C3, low C4, and normal factor B. Relevant conditions in which such a pattern could be seen include hereditary and acquired C1 inhibitor deficiency or dysfunction. Hereditary angioedema (HAE) is an autosomal dominant disease that usually presents before age 20 to 30 and is characterized by recurrent episodes of subcutaneous and/or submucosal edema involving the face, extremities, and/or abdomen. Oropharyngeal edema may be life-threatening. Pruritus and urticaria are not typically seen. There is often a family history, but 20% to 25% of cases arise from new mutations. Triggers of angioedema include trauma, infection, dental procedures, or emotional stress, with a potential increasing frequency and severity of episodes with puberty, menses, and ovulation. In type I HAE or C1 inhibitor deficiency, there is a low C1 inhibitor level and function. In type II HAE or C1 inhibitor dysfunction, the C1 inhibitor level is usually normal or high, but the C1 inhibitor function is abnormal. In type III HAE or HAE with normal C1 inhibitor, there is normal C1 inhibitor level and function, complicating the diagnosis. A subset of these patients may have mutations in factor XII, resulting in abnormal production of bradykinin.^{47–49} In all 3 situations, bradykinin has been demonstrated to be the main mediator of the angioedema. Most patients with HAE have a persistently low C4 with normal C1q and C3.⁵⁰ However, in rare cases, the C4 level can be normal between attacks.⁵¹ HAE can be differentiated from acquired C1 inhibitor deficiency (also called acquired angioedema [AAE]), by measuring C1q, which is low in AAE but not in HAE. AAE may be seen in patients with chronic lymphoproliferative disorders, such as multiple myeloma as well as rare cases of SLE that may have an autoantibody to C1 inhibitor.⁵²

Acute and chronic inflammation may lead to an increase in complement proteins, which are among the acute phase reactants synthesized by the liver in response to inflammatory cytokines.

SUMMARY

The delicate balance between complement activation and regulation thus plays a key role in innate immunity and autoimmunity. Understanding the pathways involved in complement activation and function, how the complement system is regulated, and how complement laboratory tests are interpreted will aid in the appropriate diagnosis of disorders of complement deficiency and dysregulation.

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